THE DIRECT-DECORATION ROOTED-END COMPACTIFICATION: AN EXACT ONE-SCALE MODEL FOR ROOTED COMPONENT PACKAGES

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Abstract. Let f be a quadratic polynomial in a non-principal decoration outside a directly attached satellite copy of the Mandelbrot set, and suppose its modified principal nest terminates in a primitive quadratic-like renormalization. The Kahn–Lyubich puzzle then supplies a separator piece P, an iterate $g = f^q$, two disjoint closed univalent pullbacks $Q_0, Q_1 \subset P$, and a terminal primitive root continuum K_ϵ in the first-level gap. We prove that the finite pullbacks $K_w = \phi_w(K_\epsilon)$ and the nested end fibers $E_\omega = \bigcap_m Q_{\omega|m}$ assemble into one compact planar set $\mathcal{H}_\partial \subset K(f)$ whose connected components are exactly the displayed fibers, and that the component projection $\pi_\partial \colon \mathcal{H}_\partial \to \Omega_\partial$ onto the binary address compactification is a continuous closed surjection: the component space of $\mathcal{H}_\partial$ *is* Ω_∂. Off the root, the dynamics is exact: g restricted to $\mathcal{H}_\partial \setminus K_\epsilon$ is a trivial two-sheeted covering of $\mathcal{H}_\partial$ whose sheets are the clopen sets $\mathcal{H}_\partial \cap Q_0$ and $\mathcal{H}_\partial \cap Q_1$, and π_∂ semiconjugates it to the first-symbol deletion shift. Geometric deletion of the last address symbol and dynamical deletion of the first are genuinely different parent arrows, and the root carries a separate terminal-cycle transition. We further prove that every fiber is full with connected complement; that the end compactum $\mathcal{H}_\infty = \bigcup_\omega E_\omega$ is a Cantor-indexed union of full continua realizing the one-sided binary shift by two univalent branches; that the package carries an equivariant atlas of pairwise disjoint closed Jordan gap cells; that the fibers over an *arbitrary*—possibly infinite—set of finite addresses may be replaced by larger continua inside their cells while compactness, the component structure, and (under fullness) nonseparation survive; and that finitely many protected prime-end chains at the root transport to a finite protected carrier meeting the compactum in exactly the selected fibers and preserving the transported prime ends with their cyclic order. The compactification package is natural under root enlargement within the gap, and for a full enlarged root the entire fullness and cell theory persists.

The paper is a positive control for the virtual renormalization program: it exhibits, in one concrete dynamical setting, exactly what a correctly typed rooted-end component package looks like. It does not construct a virtual Cantor package: the end fibers are not identified with small Julia sets, and no second satellite portrait, no analytic return map on an end fiber, and no uniform incidence bound is claimed.

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Date: July 27, 2026 (Draft II).

2020 *Mathematics Subject Classification.* Primary 37F10, 37F25; Secondary 37F46, 54F15.

Key words and phrases. Mandelbrot set, local connectivity, renormalization, decorations, puzzle pieces, prime ends, plane continua.

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1. Introduction

1.1. **The rooted-end problem.** The MLC conjecture—local connectivity of the Mandelbrot set—has been reduced, over three decades, to renormalization statements of increasingly combinatorial character [DH85, Hub93, Lyu97, Kah06, KL08, KL09, KKL26]. The present frontier is unbounded satellite combinatorics: traditional complex a priori bounds cannot be uniform along satellite relative periods tending to infinity [BLOT22], and the current proposals replace bounds by structural transition statements for towers of renormalizations. In Dudko's program [Dud25], the central objects are *virtual Julia sets*: connected hulls of Cantor families of small Julia sets carrying a finite part of the critical orbit. Kahn–Kapiamba–Lyubich [KKL26] construct, in their native primitive category, a positive control: a Hubbard tree inside the full dynamical carrier, lifted from the locally connected rational lamination model when the carrier itself is not locally connected.

Any such program must eventually manipulate a *rooted component package*: a distinguished root continuum, a family of homeomorphic copies of it indexed by finite dynamical addresses, limit fibers indexed by infinite addresses, transition dynamics between them, and access data (prime ends, protected collars, connectors) that survive the passage to limits. The sources used by the present program do not supply, in one theorem, a compact planar realization carrying all of the component, parent, root-arrow, and protected-access data listed here, and several plausible shortcuts are provably underdetermined (Section 8).

This paper constructs one such package completely, at one concrete scale, with every hypothesis displayed and every typing arrow realized by an actual restriction of an iterate of the polynomial. The setting is the *first direct-decoration scale* of Kahn–Lyubich [KL08, KL09]: a quadratic polynomial f chosen in a non-principal decoration outside a directly attached satellite copy of the Mandelbrot set, whose modified principal nest terminates in a primitive quadratic-like renormalization. There the puzzle supplies a separator piece P, an iterate $g = f^q$, and two disjoint closed univalent pullbacks $Q_0, Q_1 \subset P$; the nest supplies a terminal primitive root continuum K_ϵ lying in the first-level gap $P \setminus (Q_0 \cup Q_1)$.

1.2. **Main results.** Write $\phi_a \colon P \to Q_a$ $(a \in \{0,1\})$ for the inverse branches of g, $\phi_w = \phi_{a_1} \circ \cdots \circ \phi_{a_m}$ for a finite binary word $w = a_1 \cdots a_m$, and

$$Q_w = \phi_w(P), \qquad K_w = \phi_w(K_\epsilon), \qquad E_\omega = \bigcap_{m \geq 1} Q_{\omega|m} \quad (\omega \in \{0,1\}^{\mathbb{N}}).$$

The binary address compactification Ω_∂ consists of all finite words and all infinite words, topologized as a compact subset of $\{0,1,\#\}^{\mathbb{N}}$. The *rooted-end compactum* is

$$\mathcal{H}_\partial = \bigcup_w K_w \ \cup \ \bigcup_\omega E_\omega.$$

The main results, proved under the displayed Standing Hypotheses 2.2, are:

(I) *Compact component realization and quotient* (Theorems 5.3 and 5.4). $\mathcal{H}_\partial$ is a compact subset of $K(f)$ (of $J(f)$ under (H3)); its connected components are exactly the fibers K_w and E_ω; and the component projection $\pi_\partial \colon \mathcal{H}_\partial \to \Omega_\partial$ is a continuous closed surjection, so the component space of $\mathcal{H}_\partial$ is homeomorphic to Ω_∂. In particular the component decomposition is upper semicontinuous.

(II) *Exact two-branch partial shift* (Theorem 5.5). $\mathcal{H}_\partial \setminus K_\epsilon$ is the disjoint union of the clopen sheets $H_a = \mathcal{H}_\partial \cap Q_a = \phi_a(\mathcal{H}_\partial)$, and $g \colon \mathcal{H}_\partial \setminus K_\epsilon \to \mathcal{H}_\partial$ is a trivial two-sheeted covering semiconjugated by π_∂ to the first-symbol deletion shift σ_{pre}.

(III) *Two parent maps and a separate root arrow* (Remarks 5.6 and 5.7, Figure 1). Geometric nesting deletes the *last* symbol, $\mathrm{gpar}(wa) = w$; dynamical transport deletes the *first*, $\sigma_{\mathrm{pre}}(aw) = w$. These are different arrows with different domains, and the package keeps both. The root carries a third, separate transition τ_{root}.

(IV) *Fullness, nonseparation, and the end compactum* (Theorem 6.1 and Corollary 6.2). Every fiber is a full continuum, $\widehat{\mathbb{C}} \setminus \mathcal{H}_\partial$ is connected, and the end compactum $\mathcal{H}_\infty = \bigcup_\omega E_\omega$ is a compact nonseparating set whose components are the E_ω, with component space the Cantor space $\{0,1\}^{\mathbb{N}}$ and with g realizing the one-sided binary shift through two univalent branches.

(V) *Cellwise replacement at arbitrary addresses, and naturality* (Theorems 5.8, 6.4 and 6.5 and Corollary 6.7). The package admits an equivariant atlas of pairwise disjoint closed Jordan cells D_w; the fibers over an *arbitrary* set of finite addresses may be replaced by larger compact connected sets inside their cells while compactness, the exact component structure, and the continuous component quotient survive, and—when the replacements are full—nonseparation survives as well. The compactification package is natural under root enlargement within the gap, and a full enlarged root carries the entire fullness and cell theory.

(VI) *Finite protected carrier* (Theorem 7.2). Finitely many protected prime-end chains at the root transport through the inverse branches to finitely many selected fibers, and there is a full connected carrier H with $H \cap \mathcal{H}_\partial = \bigcup_{w \in F} K_w$ exactly, in whose complement the transported chains survive as prime-end chains with their cyclic order.

1.3. **What this paper is, and is not.** The abstract topology here is classical: a binary rooted tree with its ends, realized as nested disks with a Cantor limit set, is the oldest compactification in the subject. The contribution is not a new compactification theory. It is the *exact dynamical realization* of the typed package in one concrete complex-dynamical setting:

- the finite fibers are actual Julia-set pullbacks and the end fibers are actual nested puzzle intersections, in one compactum inside $K(f)$;
- both parent maps are retained and distinguished, and the dynamical one is realized by named univalent branches with time q, a recorded inverse sheet, and an exact two-sheeted covering off the root;
- the component space is computed exactly, as a continuous closed quotient;
- fullness, complement connectedness, the cell atlas, arbitrary-address replacement, and the protected carrier are proved for this realization, not for a symbolic model imposed afterwards;
- the compactification package is natural under root enlargement within the gap (Theorem 5.8 and Corollary 6.7), which is the enlargement the virtual program actually performs.

Equally important is what is *not* claimed; see Non-claims 2.3 and Section 8. This paper does not construct Dudko's virtual Cantor package. The end fibers are not identified with small Julia sets; no second satellite portrait, no analytic return map on an end fiber, and no uniform incidence bound is constructed. The honest role of the results is that of a positive control: a complete, directly visible, one-scale model showing what a correctly typed rooted-end component package looks like, against which proposed virtual constructions can be audited. The statements that remain open are recorded as Problems 8.2 and 8.3, and Appendix B records the adversarial boundary cases the proofs are required to pass.

1.4. **Organization.** Section 2 fixes the dynamical setting and states the standing hypotheses, together with the local pullback-boundary lemma that source-locks them. Section 3 collects the plane-topology toolkit (Čech–Alexander duality for disjoint full packets, nested-packet nonseparation, pointed Schoenflies transport, Jordan neighborhoods, slit and tree lemmas, and the crosscut-chain conventions). Section 4 constructs protected regulated connectors from certificates and proves the exact free-component criterion. Section 5 proves the compactification, quotient, and partial-shift theorems and the two-parent-map structure; Section 6 proves fullness, nonseparation, the end compactum, the gap-cell atlas, arbitrary-address replacement, and full-root naturality; Section 7 proves the finite protected-carrier theorem. Section 8 delimits scope and states the open problems. Appendix A records the concordance with the companion program manuscript, and Appendix B the adversarial boundary cases.

2. The first direct-decoration scale and the standing hypotheses

2.1. **Dynamical setting.** Let $f = f_c(z) = z^2 + c$ with c in a non-principal decoration attached to a directly attached satellite copy $\mathcal{M}_{t/q}$, $q > 1$, of the Mandelbrot set, with c outside the copy itself. In the Kahn–Lyubich decoration puzzle [KL08], let Y^1 denote the critical depth-one piece and Z_j^1 the off-critical depth-one pieces, and set

$$n = \min\{m \geq 1 : f^{mq}(0) \in \operatorname{int} Z_j^1 \text{ for some } j\}. \tag{2.1}$$

The minimum is finite because c lies outside the satellite copy. The terminal component Z_κ^1 and the intermediate pullback components determine the escape route, and the selected escape piece is the critical piece

$$V_{\mathrm{esc}} = f^{-nq}\big(Z_\kappa^1\big)\big|_0 = Y^{nq+1}. \tag{2.2}$$

Kahn–Lyubich define this piece as the *first kingdom* and prove that every domain of its first-return map is compactly contained in it [KL08, KL09].

We assume throughout, as part of the setting, that the modified principal nest built on (2.2) *terminates in a primitive quadratic-like renormalization*; this is the stopping condition of the Kahn–Lyubich dynasty construction [KL09]. The terminal datum is a polynomial-like first-return branch

$$f^p\colon A_0 \longrightarrow B, \qquad A_0 \Subset B, \tag{2.3}$$

of some period p, where A_0 is the critical first-return domain containing 0 [KL09, DH85]. Let K_ϵ denote its filled Julia set (the *terminal primitive root*) and let $\mathcal{K}_i = f^i(K_\epsilon)$, $0 \le i < p$, denote the terminal cycle, so $\mathcal{K}_0 = K_\epsilon$.

Proposition 2.1 (Critical-base cycle exclusion)**.** *For $1 \le i < p$ we have $\mathcal{K}_i \cap B = \varnothing$. The complete cycle itinerary is $\mathcal{K}_1 \to \mathcal{K}_2 \to \cdots \to \mathcal{K}_{p-1} \to \mathcal{K}_0$, and the chronological degree word of the cycle, starting at $\mathcal{K}_0$, is $(2, 1, \dots, 1)$.*

Proof. Equation (2.3) is a first-return branch, so $f^i(A_0) \cap B = \varnothing$ for $1 \le i < p$. Since $K_\epsilon \subset A_0$, the same holds for $\mathcal{K}_i = f^i(K_\epsilon)$. Primitivity makes the $\mathcal{K}_i$ pairwise disjoint. The only critical point is $0 \in \mathcal{K}_0$, because A_0 is the critical first-return domain; an intermediate critical encounter would make some $\mathcal{K}_i$, $1 \le i < p$, contain $0 \in B$, contradicting the first display. Hence only the first factor of the cycle has degree two. □

2.2. **The package data.** The compactification uses the following data extracted from the construction above. Let

$$P = Y^{(n-1)q+1}, \qquad g = f^q, \qquad g\colon Q_a \longrightarrow P \quad (a \in \mathtt{A} := \{0,1\}) \tag{2.4}$$

where Q_0, Q_1 are the two univalent pullbacks of P under g inside P supplied by the decoration construction [KL08]: each Q_a is the closure of a component of $g^{-1}(\operatorname{int} P)$, and the two components are distinct. Let $V^0 = Y^{nq+1}$ be the first kingdom (2.2), so that $K_\epsilon \Subset \operatorname{int} V^0$ and V^0, Q_0, Q_1 are distinct puzzle pieces of the common depth $nq+1$. Write $\phi_a\colon P \to Q_a$ for the inverse branches.

We isolate exactly the properties of these data that the proofs use.

Standing Hypotheses 2.2. Throughout the paper:

(H1) P, Q_0, Q_1 are *closed* Jordan disks with $Q_0 \cup Q_1 \subset P$, and the two closed pieces are disjoint: $Q_0 \cap Q_1 = \varnothing$. Disjointness of the closed pieces, not merely of their interiors, is used.

(H2) Each restriction $g|_{Q_a}$ is proper of degree one and extends to a homeomorphism $g\colon Q_a \to P$ of the closed Jordan pieces; equivalently, each $\phi_a\colon P \to Q_a$ is an orientation-preserving homeomorphism of closed Jordan disks.

(H3) $K(f) = J(f)$: the filled Julia set of f has empty interior. This is an explicit assumption; it holds, for instance, whenever f has no attracting and no neutral cycle, and in particular for every infinitely renormalizable parameter.

(H4) The root K_ϵ is a nondegenerate full continuum (a full continuum containing more than one point) with

$$K_\epsilon \subset \operatorname{int} P \setminus (Q_0 \cup Q_1); \tag{2.5}$$

it lies in the *first-level gap*. We write G_0 for the component of $\operatorname{int} P \setminus (Q_0 \cup Q_1)$ containing K_ϵ.

Non-claims 2.3. Equally binding throughout:

(N1) The root transition $\tau_{\mathrm{root}}\colon K_\epsilon \mapsto f^q(K_\epsilon)$ is separate data (Remark 5.7); it is *not* an extension of the first-symbol deletion map σ_{pre}, and no assertion of permanent escape from $\mathcal{H}_\partial$ is made.
(N2) No second satellite portrait is constructed, and the end fibers E_ω are *not* identified with small Julia sets.
(N3) No analytic return map *on* an end fiber is constructed: no first-return or polynomial-like self-return structure on any single E_ω is claimed. The analytic transitions $E_{a\omega} \to E_\omega$ of (5.5) *are* proved; they are a different assertion.
(N4) No uniform (period-independent) incidence or marking bound is claimed; see Proposition 8.1.

2.3. **Source-locking the hypotheses.** Closed-piece disjointness in (H1) is not formal vocabulary: closed puzzle pullbacks may in general touch at boundary points, as Kahn–Lyubich themselves caution [KL08], and any common point of Q_0 and Q_1 necessarily lies on both boundaries. The next lemma is the local statement that makes the exclusion argument exact.

Lemma 2.4 (Local uniqueness of a regular closed pullback)**.** *Let h be a holomorphic map defined near $z \in \mathbb{C}$ with $h'(z) \neq 0$, and let P be a closed Jordan disk containing $h(z)$. Then z belongs to the closure of at most one component of $h^{-1}(\operatorname{int} P)$.*

Proof. Choose an open topological disk V about $h(z)$ and a neighborhood W of z such that $h\colon W \to V$ is a homeomorphism. If $h(z) \in \operatorname{int} P$, shrink V so that $V \subset \operatorname{int} P$; then $V \cap \operatorname{int} P = V$ is connected. If $h(z) \in \partial P$, apply the Schoenflies theorem to a chart in which ∂P near $h(z)$ is a straight segment and shrink V to a round disk in that chart; then $V \cap \operatorname{int} P$ is an open half-disk, again connected. In both cases

$$W \cap h^{-1}(\operatorname{int} P) = (h|_W)^{-1}\big(V \cap \operatorname{int} P\big)$$

is connected. If a component C of $h^{-1}(\operatorname{int} P)$ has $z \in \overline{C}$, then C meets the connected set $W \cap h^{-1}(\operatorname{int} P)$, which is therefore contained in C. Two distinct components would both contain this nonempty connected set, contradicting disjointness of components. □

Lemma 2.5 (Realization of the standing hypotheses)**.** *The decoration data (2.4) satisfy (H1), (H2), and (H4).*

Proof. The pieces P, Q_0, Q_1 are closed Jordan puzzle pieces and the two pullbacks are closures of distinct components of $g^{-1}(\operatorname{int} P)$ [KL08]. Suppose the closed pieces Q_0, Q_1 met at a point z. Since the interiors of distinct depth-$(nq+1)$ pieces are disjoint complementary components of the depth-$(nq+1)$ puzzle graph, z lies on that graph. We claim z is a regular point of $g = f^q$. Indeed, the critical points of g are the points ζ with $f^i(\zeta) = 0$ for some $0 \le i < q$, and f maps the depth-d puzzle graph into the depth-$(d-1)$ graph; if z were critical, $0 = f^i(z)$ would therefore lie on a puzzle graph. But the critical pieces Y^d are defined at every depth occurring in the construction—the critical orbit avoids the puzzle graphs [KL08]—so 0 lies in $\operatorname{int} Y^d$ and never on a graph. Hence z is a regular point of g, and Lemma 2.4 puts z in the closure of at most one component of $g^{-1}(\operatorname{int} P)$. Since $z \in Q_0 \cap Q_1$ and the Q_a are closures of distinct components, this is a contradiction. This proves (H1).

A proper degree-one holomorphic map between Jordan domains is a conformal isomorphism of the interiors; since the pieces are closed Jordan disks bounded by finitely many ray and equipotential arcs, Carathéodory extension gives a homeomorphism of the closed pieces, orientation-preserving because the interior map is holomorphic. This proves (H2).

For (H4): the filled Julia set of the polynomial-like restriction (2.3) is a full continuum [DH85]. Nondegeneracy is automatic: K_ϵ is the filled Julia set of a degree-two polynomial-like branch, whose Julia set is a perfect set [DH85], so K_ϵ is not a point. The three pieces

V^0, Q_0, Q_1 are distinct puzzle pieces of the common depth $nq+1$; the interior of each is a complementary component of the depth-$(nq+1)$ puzzle graph, so $\operatorname{int} V^0$ is disjoint from the closed pieces Q_0 and Q_1. Since $K_\epsilon \Subset \operatorname{int} V^0$ and $V^0 \subset P$ (deeper critical pieces are nested), we get (2.5). □

Remark 2.6 (Source audit and a documented fallback). The proof of (H1) rests on two source facts: the two f^q-pullbacks of the separator are distinct pieces, and the critical orbit avoids the puzzle graphs. Should a source expert question closed-piece disjointness for this pair, there is a robust fallback: Kahn–Lyubich construct, at the deeper f^{nq}-level, one univalent pullback Q^v of P for every vertex v of the separator, and state that these are *pairwise disjoint* [KL08]. Choosing two of them and setting $g = f^{nq}$ satisfies (H1)–(H2) verbatim and leaves the entire architecture of this paper intact; only the transition time per address symbol changes from q to nq. We record this as a fallback and do not silently depend on it.

Remark 2.7 (Where each hypothesis is used). (H1) and (H4) drive every separation statement (Lemma 5.1). (H2) is used three ways: to push the nested intersection through g (Theorem 5.3), to transport fullness and Fill by the pointed Schoenflies extension (Lemma 3.3 and Theorem 6.1), and to transport cells, ports, and prime-end chains (Theorems 6.4 and 7.2). (H3) is used exactly once, to upgrade $\mathcal{H}_\partial \subset K(f)$ to $\mathcal{H}_\partial \subset J(f)$ (Theorem 5.3); every other statement is independent of it.

3. Plane-topology toolkit

For a compact set $K \subset \mathbb{C}$ let

$$\operatorname{Fill}(K) = K \cup \bigcup \{\text{bounded components of } \mathbb{C} \setminus K\}. \tag{3.1}$$

A compact connected set $K \subsetneq S^2$ is *full* if $S^2 \setminus K$ is connected. For bounded $K \subset \mathbb{C}$, viewing $\mathbb{C} \subset \widehat{\mathbb{C}} = S^2$, this is equivalent to $\operatorname{Fill}(K) = K$; the spherical formulation is the one used below, since the packets will include the exterior continuum $S^2 \setminus U$ of a Jordan domain.

The first lemma is the Čech form of Alexander duality [Hat02, pp. 256–257]: for every nonempty proper compact $K \subset S^2$,

$$\widetilde{H}_0(S^2 \setminus K; \mathbb{Z}) \;\cong\; \check{H}^1(K; \mathbb{Z}). \tag{3.2}$$

The Čech form, rather than the locally contractible special case, is required: the compacta below are Julia-set pullbacks with no local structure assumptions.

Lemma 3.1 (Finite disjoint full packets). *If $K_0, \dots, K_N \subset S^2$ are pairwise disjoint nonempty full continua, then $S^2 \setminus \bigcup_{r=0}^{N} K_r$ is connected.*

Proof. Fullness of K_r makes $S^2 \setminus K_r$ connected, so (3.2) gives $\check{H}^1(K_r; \mathbb{Z}) = 0$. The finitely many pairwise disjoint compacta have pairwise disjoint open neighborhoods; the covers of $\bigcup_r K_r$ subordinate to these neighborhoods are cofinal among all open covers, and each has nerve equal to the disjoint union of the individual nerves. Passing to first cohomology and then to the direct limit defining Čech cohomology,

$$\check{H}^1\Big(\bigcup_r K_r; \mathbb{Z}\Big) \cong \bigoplus_r \check{H}^1(K_r; \mathbb{Z}) = 0.$$

A second application of (3.2) makes the complement connected. □

Lemma 3.2 (Nested finite-full-packet nonseparation). *Let $F_0 \supset F_1 \supset \cdots$ be compact proper subsets of S^2, each a finite union of pairwise disjoint full continua. Then $\bigcap_m F_m$ has connected complement in S^2. If moreover U is a Jordan domain with $F_m \Subset U$ for every m and each*

$F_m \cup (\widehat{\mathbb{C}} \setminus U)$ is again a finite union of pairwise disjoint full continua, then $U \setminus \bigcap_m F_m$ is connected.

Proof. By Lemma 3.1 every $S^2 \setminus F_m$ is connected and nonempty. These complements increase with m, all contain a common point (any point of $S^2 \setminus F_0$), and their union is the complement of the intersection; an increasing union of connected sets with a common point is connected. For the localized statement, apply the same argument to the packets $F_m \cup (\widehat{\mathbb{C}} \setminus U)$: the complement of the intersection is then exactly $U \setminus \bigcap_m F_m$. □

Lemma 3.3 (Pointed Schoenflies transport)**.** *Let $\phi\colon D \to D'$ be an orientation-preserving homeomorphism of closed Jordan disks in $\mathbb{C}$. Then ϕ extends to an orientation-preserving homeomorphism Φ of $S^2 = \widehat{\mathbb{C}}$ with $\Phi(\infty) = \infty$. Consequently, for every compact $S \subset D$:*

(a) *if S is a full continuum, so is $\phi(S)$; and*
(b) *$\Phi(\mathrm{Fill}(S)) = \mathrm{Fill}(\phi(S))$; in particular $\mathrm{Fill}(\phi(S)) = \phi(\mathrm{Fill}(S))$ whenever $\mathrm{Fill}(S) \subset D$.*

Proof. The complements $\Delta = S^2 \setminus \operatorname{int} D$ and $\Delta' = S^2 \setminus \operatorname{int} D'$ are closed Jordan disks in S^2 containing ∞ in their interiors. By the Schoenflies theorem choose homeomorphisms $h\colon \Delta \to \overline{\mathbb{D}}$ and $h'\colon \Delta' \to \overline{\mathbb{D}}$; composing with self-homeomorphisms of $\overline{\mathbb{D}}$ that fix the boundary pointwise, arrange $h(\infty) = h'(\infty) = 0$. The circle homeomorphism $\beta = h' \circ (\phi|_{\partial D}) \circ h^{-1}$ extends by coning, $\widehat{\beta}(re^{i\theta}) = r\,\beta(e^{i\theta})$, to a homeomorphism of $\overline{\mathbb{D}}$ fixing 0. Define $\Phi = \phi$ on D and $\Phi = (h')^{-1} \circ \widehat{\beta} \circ h$ on Δ; the two definitions agree on ∂D, so Φ is a sphere homeomorphism, and $\Phi(\infty) = (h')^{-1}(0) = \infty$. It is orientation-preserving because it is so on the open disk $\operatorname{int} D$.

For (b): Φ carries the components of $\widehat{\mathbb{C}} \setminus S$ bijectively onto the components of $\widehat{\mathbb{C}} \setminus \phi(S)$ and matches the component of ∞ with the component of ∞; hence it carries the union of S with its bounded complementary components onto the corresponding union for $\phi(S)$. When $\mathrm{Fill}(S) \subset D$, the restriction of Φ to $\mathrm{Fill}(S)$ is ϕ itself, giving the displayed identity with ϕ. (a) is the special case in which S has no bounded complementary components: then neither does $\phi(S)$, and $S^2 \setminus \phi(S) = \Phi(S^2 \setminus S)$ is connected. □

Lemma 3.4 (Full-continuum Jordan neighborhood)**.** *If a nondegenerate full continuum K is compactly contained in an open set $G \subset \mathbb{C}$, there is a closed Jordan disk D with*

$$K \subset \operatorname{int} D \subset D \Subset G. \tag{3.3}$$

Proof. Uniformize the simply connected exterior by $\psi\colon \{|z| > 1\} \to \widehat{\mathbb{C}} \setminus K$. The filled interiors of the analytic Jordan curves $\psi(\{|z| = R\})$, $R > 1$, are closed Jordan disks decreasing to K as $R \downarrow 1$. If none lay in G, compactness would place a point of the closed set $\widehat{\mathbb{C}} \setminus G$ in their intersection K, a contradiction. □

Lemma 3.5 (One-point tree attachments are full)**.** *If a finite embedded plane tree meets each of finitely many pairwise disjoint full continua in exactly one point and otherwise misses them, the union is a full continuum.*

Proof. A finite embedded plane tree is nonseparating. Attach the full continua one at a time. At each step the two nonseparating continua meet in a connected singleton, and the Janiszewski separation lemma implies that their union is nonseparating. Induction proves that the final compact connected set has connected complement. (Equivalently: connectedness of each intersection and vanishing of the first Čech cohomology of the two pieces force, by Mayer–Vietoris and (3.2), vanishing first Čech cohomology of the union.) □

Lemma 3.6 (One-ended slit lemma)**.** *Let $\Omega \subset \mathbb{C}$ be a domain and let $\delta_1, \dots, \delta_s \colon [0,1] \to \overline{\Omega}$ be pairwise disjoint Jordan arcs with $\delta_i(0) \in \partial\Omega$ and $\delta_i((0,1]) \subset \Omega$. Then*

$$\Omega \setminus \bigcup_{i=1}^{s} \delta_i((0,1])$$

is path connected. Moreover,

$$\Omega \setminus \bigcup_{i=1}^{s} \delta_i((0,1))$$

is path connected and contains every free tip $\delta_i(1)$.

Proof. For the first statement, treat one slit δ first. A polygonal path between two points of the complement is compactly contained in Ω, hence misses an initial tail $\delta((0,\varepsilon))$. Perturb the path to meet the compact arc $\delta([\varepsilon,1])$ transversely in finitely many points. In a thin topological rectangle around that arc, replace every crossing by travel along the two long sides; the sides join around the free endpoint $\delta(1)$. (Planar arcs are tame [Moi77], so such rectangle neighborhoods exist; the same fact underlies every detour argument of this kind.) The replacement stays in Ω and avoids the whole slit. The removed slit is relatively closed in Ω, so its complement is again a domain, and induction over the finite disjoint family proves the claim.

For the second statement, write $O = \Omega \setminus \bigcup_i \delta_i((0,1])$, which is open, path connected by the first part, and misses every tip. Fix a tip $p_i = \delta_i(1)$. By injectivity and compactness there is $t_i < 1$ with $\operatorname{dist}(p_i, \delta_i([0,t_i])) > 0$; choose an open disk $B_i \ni p_i$ with $\overline{B_i} \subset \Omega$, meeting no δ_j with $j \neq i$, missing $\delta_i([0,t_i])$, and with the B_i pairwise disjoint. By tameness of the arc $A_i = \delta_i([t_i,1])$ [Moi77], a homeomorphism h of the plane carries A_i to the segment $[0,1] \times \{0\}$ with $h(p_i) = (1,0)$. For small $s > 0$ the segment $\{(1+\sigma, 0) : 0 \le \sigma \le s\}$ lies in the open set $h(B_i)$ and, except for its initial point, misses $h(A_i)$; its preimage is an arc from p_i into B_i whose interior misses every slit, hence lies in O. Thus every tip is joined to the path-connected set O by an arc meeting the slits only at the tip itself, and $O \cup \{p_1, \dots, p_s\}$ is path connected. □

3.1. **Crosscut chains.** Finally we fix the crosscut conventions. A *crosscut* of the complement of a full continuum A is a Jordan arc whose interior lies in $\widehat{\mathbb{C}} \setminus A$ and whose two distinct endpoints lie on ∂A. A *crosscut chain* $(\gamma_n, D_n)_{n \ge 1}$ is a sequence of pairwise disjoint crosscuts γ_n together with selected sides D_n, where:

(C1) D_n is a selected component of $\widehat{\mathbb{C}} \setminus (A \cup \gamma_n)$;
(C2) the chain is *nulled*:

$$\overline{D_{n+1}} \setminus A \subset D_n \qquad (n \ge 1); \tag{3.4}$$

(C3) $\operatorname{diam} \gamma_n \to 0$.

Condition (3.4) contains the usual null-chain relation: since the open arc γ_{n+1}° lies in $\overline{D_{n+1}} \setminus A$, the later crosscuts satisfy $\gamma_{n+1}^{\circ} \subset D_n$, so for $n \ge 2$ every crosscut interior lies in D_1. The chain defines a prime end of $\widehat{\mathbb{C}} \setminus A$ with impression $\bigcap_n \overline{D_n}$. We call $\overline{D_n} \setminus A$ the *relative collar tail* at level n. Relative collar tails of different chains may have closures meeting inside A—in particular, overlapping prime-end impressions are allowed, and no local connectivity of A is ever assumed.

Lemma 3.7 (Protected crosscut survival and order)**.** *Let A be a full continuum, let $H \supset A$ be a full continuum, and let (γ_n, D_n) be a crosscut chain of $\widehat{\mathbb{C}} \setminus A$. If*

$$(H \setminus A) \cap (\overline{D_1} \setminus A) = \varnothing, \tag{3.5}$$

then the crosscuts $(\gamma_n)_{n\geq 2}$ with their selected sides form a prime-end chain in $\widehat{\mathbb{C}} \setminus H$; the endpoints remain boundary points and the new selected sides contain the old D_n. A finite family of such chains with pairwise disjoint relative terminal collar tails induces distinct prime ends of $\widehat{\mathbb{C}} \setminus H$ with unchanged cyclic order. Here the preserved cyclic order is that of the transported crosscut family in the one ambient complement $\widehat{\mathbb{C}} \setminus H$; no identification of prime-end circles across different atoms is asserted.

Proof. For $n \geq 2$, (3.4) gives $\gamma_n^\circ \subset D_{n-1} \subset D_1$, so $\gamma_n^\circ \subset \overline{D_1} \setminus A$ misses H by (3.5), while the two distinct endpoints of γ_n lie in $A \subset H$. Every neighborhood of either endpoint meets the crosscut interior, so both endpoints remain on the boundary of the enlarged complement; thus γ_n remains a crosscut of $\widehat{\mathbb{C}} \setminus H$. A path joining D_n to the opposite side of γ_n without meeting $H \cup \gamma_n$ would give the same forbidden path in $\widehat{\mathbb{C}} \setminus A$; hence the selected sides remain nested, and the unchanged crosscut diameters still tend to zero.

Equivalent chains eventually divide one another; pairwise disjoint relative terminal collar tails prevent that, so the induced prime ends are distinct. For a finite family, choose one deep oriented crosscut from each chain: prime-end cyclic order is the separation order of these crosscuts, and neither the crosscuts nor their selected sides changed, so every ordered triple retains its orientation. □

4. Protected regulated connectors

This section constructs the connector technology consumed by the protected-carrier theorem in Section 7. Throughout, U is a Jordan domain, $A_1, \dots, A_s \Subset U$ are pairwise disjoint full continua (the *atoms*), and $Z \Subset U \setminus \bigcup_i A_i$ is a compact *forbidden set*. For every protected access to A_i a crosscut chain $\Gamma_{i,a} = (\gamma_{i,a,n}, D_{i,a,n})_{n\geq 1}$ of $\widehat{\mathbb{C}} \setminus A_i$ is chosen, truncated so that the relative terminal collar tails $\overline{D_{i,a,1}} \setminus A_i$ are pairwise disjoint, lie in U, and miss Z and the other atoms. The closures of two tails may meet inside their common atom: overlapping prime-end impressions are allowed.

Definition 4.1 (Protected ribbon-routing certificate)**.** A *protected ribbon-routing certificate* consists of the preceding data, pairwise disjoint *free-port arcs* α_i satisfying

$$\alpha_i \cap A_i = \{x_i\} = \{\alpha_i(0)\}, \qquad \alpha_i((0,1]) \cap \Big(Z \cup \bigcup_j A_j \cup \bigcup_{j,a} \big(\overline{D_{j,a,1}} \setminus A_j\big)\Big) = \varnothing,$$

and an embedded finite tree $G \subset U$ joining the outer port endpoints $\alpha_i(1)$ while missing the same protected and forbidden sets, subject to the incidence conditions

$$G \cap \alpha_i = \{\alpha_i(1)\} \quad (1 \leq i \leq s), \qquad T_{\mathfrak{R}} := G \cup \bigcup_i \alpha_i \text{ is a finite embedded plane tree.} \quad (4.1)$$

The planar cyclic order of half-edges at every vertex of $T_{\mathfrak{R}}$ is retained. The free port points x_i are required not to be impressions of the protected chains.

Given a certificate $\mathfrak{R}$, define the *carrier*

$$H_{\mathfrak{R}} = T_{\mathfrak{R}} \cup \bigcup_i A_i = G \cup \bigcup_i \alpha_i \cup \bigcup_i A_i. \quad (4.2)$$

The *attachment trace* of a carrier $H \supset A_i$ at the atom A_i is the compact set $\overline{H \setminus A_i} \cap A_i$.

Theorem 4.2 (Protected regulated connector realization)**.** *Every protected ribbon-routing certificate has an actual carrier $H_{\mathfrak{R}}$ with the following properties:*

(i) *$H_{\mathfrak{R}} \Subset U$ is compact, connected, and full;*

(ii) *it contains every A_i, misses Z, and has singleton attachment trace: $\overline{H_{\mathfrak{R}} \setminus A_i} \cap A_i = \{x_i\}$ for every i;*
(iii) *every protected crosscut chain remains a prime-end chain in $\widehat{\mathbb{C}} \setminus H_{\mathfrak{R}}$;*
(iv) *the survival map on the protected prime ends is injective and preserves the cyclic order of the transported family in $\widehat{\mathbb{C}} \setminus H_{\mathfrak{R}}$;*
(v) *the connector-created sides carry the boundary order of the retained ribbon tree; and*
(vi) *no local connectivity of any A_i is required.*

Proof. By (4.1), $T_{\mathfrak{R}}$ is a finite embedded plane tree, and it meets each atom A_i exactly in the singleton $\{x_i\}$: the port α_i meets A_i exactly there, the other ports miss A_i by the certificate, and G misses every atom. Lemma 3.5 applied to $T_{\mathfrak{R}}$ and the atoms gives (i) for the union (4.2), and the certificate gives the containment and avoidance parts of (ii). For the attachment trace, $H_{\mathfrak{R}} \setminus A_i$ is the union of $T_{\mathfrak{R}} \setminus \{x_i\}$ with the other atoms; its closure adds only x_i, and the other atoms are closed and disjoint from A_i, so $\overline{H_{\mathfrak{R}} \setminus A_i} \cap A_i = \{x_i\}$.

For a protected chain at A_i, the set $H_{\mathfrak{R}} \setminus A_i$ misses the relative tail $\overline{D_{i,a,1}} \setminus A_i$ by the certificate, verifying (3.5); Lemma 3.7 then proves (iii) and (iv), including the stated reading of cyclic order. A small regular neighborhood of the finite tree $T_{\mathfrak{R}}$ traverses every edge on two sides in exactly the cyclic boundary word of its ribbon structure, giving (v). The argument uses crosscut chains only, so no Euclidean boundary extension of an atom is assumed, giving (vi). □

The existence of a certificate is characterized by a free-component condition. Freeze the inventory: let Q be the union of all atoms, all relative terminal collar tails $\overline{D_{i,a,1}} \setminus A_i$, and the forbidden set Z, and put

$$\Omega_{\text{free}} = U \setminus Q. \tag{4.3}$$

A component Ω_0 of Ω_{free} has *simultaneous singleton ports* if there are pairwise disjoint arcs α_i with $\alpha_i \cap A_i = \{\alpha_i(0)\} = \{x_i\}$, x_i not an impression of a protected chain, and $\alpha_i((0,1]) \subset \Omega_0$.

Theorem 4.3 (Protected free-component criterion)**.** *A protected ribbon-routing certificate exists if and only if one component of Ω_{free} has simultaneous singleton ports to every required atom.*

Proof. Necessity: a certificate's tree $T_{\mathfrak{R}}$ with its atoms deleted is connected and lies in Ω_{free}, so all its port interiors lie in one component.

Sufficiency: choose the simultaneous port arcs in Ω_0. Their interiors $\alpha_i((0,1])$ are finitely many disjoint one-ended slits of the domain Ω_0, with $\alpha_i(0) = x_i \in \partial\Omega_0$. By the Moreover clause of Lemma 3.6, the set

$$O^+ = \Omega_0 \setminus \bigcup_i \alpha_i((0,1))$$

is path connected and contains every outer tip $\alpha_i(1)$. Join the finitely many tips pairwise by paths in O^+; since $O^+ \setminus \{\text{tips}\} = \Omega_0 \setminus \bigcup_i \alpha_i((0,1])$ is open, the path interiors may be perturbed within that open set into general position, avoiding the finitely many tips except at their own endpoints, and then subdivided at their finitely many crossings. An inclusion-minimal connected subgraph G spanning the tip vertex set is a finite embedded tree with $G \cap \alpha_i = \{\alpha_i(1)\}$ and $G \cap \bigcup_i \alpha_i((0,1)) = \varnothing$. Attaching each port arc α_i to G at the single point $\alpha_i(1)$—the ports being pairwise disjoint and meeting G nowhere else—yields a finite embedded plane tree $T_{\mathfrak{R}} = G \cup \bigcup_i \alpha_i$, so (4.1) holds and the data form a protected ribbon-routing certificate. □

5. The rooted-end compactification

5.1. **Addresses and fibers.** Let $\mathtt{A} = \{0, 1\}$, let $\mathtt{A}^{<\mathbb{N}}$ denote the finite binary words including the empty word ϵ, and let $\mathtt{A}^{\mathbb{N}} = \mathtt{A}^{\mathbb{N}}$ denote the ends. For $w = a_1 \cdots a_m \in \mathtt{A}^{<\mathbb{N}}$ put

$$\phi_w = \phi_{a_1} \circ \cdots \circ \phi_{a_m}, \qquad Q_w = \phi_w(P), \qquad K_w = \phi_w(K_\epsilon), \tag{5.1}$$

with $\phi_\epsilon = \mathrm{id}$, $Q_\epsilon = P$, $K_\epsilon = K_\epsilon$. For $\omega \in \mathtt{A}^{\mathbb{N}}$ put

$$E_\omega = \bigcap_{m \geq 1} Q_{\omega|m}. \tag{5.2}$$

The *binary address compactification* is

$$\Omega_\partial = \{w\#^\infty : w \in \mathtt{A}^{<\mathbb{N}}\} \cup \mathtt{A}^{\mathbb{N}} \ \subset \ \{0, 1, \#\}^{\mathbb{N}}, \tag{5.3}$$

with the product topology; we identify w with $w\#^\infty$. The *rooted-end compactum* is

$$\mathcal{H}_\partial = \bigcup_{w \in \mathtt{A}^{<\mathbb{N}}} K_w \ \cup \ \bigcup_{\omega \in \mathtt{A}^{\mathbb{N}}} E_\omega. \tag{5.4}$$

Write $\mathrm{fib}(w) = K_w$ and $\mathrm{fib}(\omega) = E_\omega$ for the realization map on Ω_∂.

Two words are *comparable* if one is a prefix of the other (written $u \sqsubseteq v$), and an end ω *extends* u if $\omega|_{|u|} = u$.

Lemma 5.1 (Cylinder separations)**.** *Under Standing Hypotheses 2.2:*

(a) $Q_{wa} = \phi_w(Q_a) \subset Q_w$*; appending symbols descends through nested closed Jordan disks, and* $g^{|w|} \colon Q_w \to P$ *is a homeomorphism.*
(b) *If* u, v *are incomparable, then* $Q_u \cap Q_v = \varnothing$*.*
(c) *If* u *is a strict prefix of* v*, then* $K_u \cap Q_v = \varnothing$*. In particular* K_u *is disjoint from* K_v *and from every* E_ω *with* ω *extending* u*.*
(d) *Distinct fibers of the realization map are pairwise disjoint.*

Proof. (a) is immediate from (H2) and $\phi_{wa} = \phi_w \circ \phi_a$. For (b), write $u = cas$, $v = ca't$ with $a \neq a'$ for the longest common prefix c; then $Q_u \subset Q_{ca} = \phi_c(Q_a)$ and $Q_v \subset \phi_c(Q_{a'})$, and these are disjoint because $Q_a \cap Q_{a'} = \varnothing$ by (H1) and ϕ_c is injective. For (c), write $v = uat$; then $Q_v \subset \phi_u(Q_a)$ while $K_u = \phi_u(K_\epsilon)$ with $K_\epsilon \cap Q_a = \varnothing$ by (H4), and injectivity of ϕ_u separates them. (d) combines (b) and (c): incomparable addresses have disjoint cylinders at the first difference, and comparable distinct addresses fall under (c) (every end fiber E_ω lies in $Q_{\omega|m}$ for all m, and two distinct ends are incomparable at some depth). □

Lemma 5.2 (Label-limit lemma)**.** *Let* $\alpha_j \to \alpha$ *in* Ω_∂ *and let* $x_j \in \mathrm{fib}(\alpha_j)$ *with* $x_j \to x$ *in* $\mathbb{C}$*. Then* $x \in \mathrm{fib}(\alpha)$*.*

Proof. If $\alpha = w\#^\infty$ is a finite address, it is isolated in Ω_∂: any address agreeing with it through position $|w| + 1$ (that is, on w followed by #) equals it. Hence $\alpha_j = \alpha$ eventually, and $x \in K_w$ because K_w is closed. If $\alpha = \omega \in \mathtt{A}^{\mathbb{N}}$, fix m. Eventually α_j agrees with ω on the first m symbols; whether α_j is a finite word $v \sqsupseteq \omega|m$ or an end extending $\omega|m$, its fiber lies in the closed cylinder $Q_{\omega|m}$ by Lemma 5.1(a). Hence $x \in Q_{\omega|m}$ for every m, that is, $x \in E_\omega$. □

5.2. **The compactification theorem.**

Theorem 5.3 (Direct-decoration rooted-end compactification)**.** *Under Standing Hypotheses 2.2:*

(i) Ω_∂ *is compact; the finite addresses are isolated, and finite words of increasing length converge to an end exactly by eventual agreement on every fixed prefix.*

(ii) *$\mathcal{H}_\partial$ is a compact subset of $K(f)$; under (H3), $\mathcal{H}_\partial \subset J(f)$. Every fiber (5.1), (5.2) is a nonempty continuum.*
(iii) *The connected components of $\mathcal{H}_\partial$ are exactly the fibers K_w ($w \in \mathtt{A}^{<\mathbb{N}}$) and E_ω ($\omega \in \mathtt{A}^{\mathbb{N}}$).*
(iv) *The realization $\Omega_\partial \ni \alpha \mapsto \mathrm{fib}(\alpha)$ has closed graph; equivalently, it is upper semicontinuous as a map into the compact subsets of $\mathcal{H}_\partial$.*
(v) *The dynamics acts through actual univalent branches of f^q:*

$$g(K_{aw}) = K_w, \qquad g(E_{a\omega}) = E_\omega \qquad (a \in \mathtt{A}), \tag{5.5}$$

each transition having time q and inverse sheet the deleted first symbol a.

Proof. (i). The product $\{0,1,\#\}^{\mathbb{N}}$ is compact. The subset (5.3) is defined by the closed conditions "if the k-th symbol is # then so is the $(k+1)$-st": each violating set is a clopen cylinder, so Ω_∂ is closed, hence compact. Isolation of finite addresses and the stated convergence are immediate in the product topology (see the proof of Lemma 5.2).

(ii). Each Q_w is a closed Jordan disk and $g^{|w|}\colon Q_w \to P$ is a homeomorphism (Lemma 5.1(a)), so each $K_w = \phi_w(K_\epsilon)$ is a continuum, and each E_ω is a nested intersection of continua, hence a nonempty continuum. Every $z \in E_\omega$ satisfies $g^m(z) \in P$ for all $m \geq 0$; the intermediate iterates $f^{mq+r}(z)$, $0 \leq r < q$, lie in the compact union $\bigcup_{r=0}^{q-1} f^r(P)$, so the forward orbit of z is bounded and $z \in K(f)$. The root satisfies $K_\epsilon \subset K(f)$: its points have f^p-orbits in A_0, and the intermediate f-iterates range over the compact sets $f^i(A_0)$, $i < p$. Complete invariance of $K(f)$ then puts every $K_w = \phi_w(K_\epsilon) \subset f^{-q|w|}(K(f)) = K(f)$. Under (H3), $K(f) = J(f)$.

For compactness of $\mathcal{H}_\partial$, let $x_j \in \mathcal{H}_\partial$ converge to x, and let $\alpha_j \in \Omega_\partial$ be addresses with $x_j \in \mathrm{fib}(\alpha_j)$. By compactness of Ω_∂ pass to a subsequence with $\alpha_j \to \alpha$; Lemma 5.2 gives $x \in \mathrm{fib}(\alpha) \subset \mathcal{H}_\partial$.

(iv). The same argument applied without passing to a subsequence of the x_j proves that the graph $\{(\alpha, x) : x \in \mathrm{fib}(\alpha)\}$ is closed in $\Omega_\partial \times \mathcal{H}_\partial$. Since $\mathcal{H}_\partial$ is compact, a closed graph is equivalent to upper semicontinuity of the set-valued realization.

(iii). Each fiber is connected by (ii), and distinct fibers are disjoint by Lemma 5.1(d). It remains to show that no connected subset of $\mathcal{H}_\partial$ meets two fibers. For a finite word u, consider

$$\mathcal{H}_\partial[u] := \bigcup_{v \sqsupseteq u} K_v \cup \bigcup_{\omega \sqsupseteq u} E_\omega.$$

We claim $\mathcal{H}_\partial[u]$ is clopen in $\mathcal{H}_\partial$. First, $\mathcal{H}_\partial[u] = \mathcal{H}_\partial \cap Q_u$: fibers with address extending u lie in Q_u by Lemma 5.1(a), fibers with address incomparable to u miss Q_u by (b), and fibers K_v with v a strict prefix of u miss Q_u by (c). Hence $\mathcal{H}_\partial[u]$ is closed. Second, its complement $\mathcal{H}_\partial \setminus \mathcal{H}_\partial[u]$ is the union of the fibers over the addresses not extending u; these addresses form a clopen, hence compact, subset $N_u \subset \Omega_\partial$, and the union of fibers over any compact address set is closed: the graph of the realization restricted to N_u is a closed subset of the compact space $N_u \times \mathcal{H}_\partial$ ((iv)), hence compact, and its projection to $\mathcal{H}_\partial$ is therefore compact. Thus $\mathcal{H}_\partial[u]$ is clopen.

Now let $C \subset \mathcal{H}_\partial$ be connected and meet the fibers of two distinct addresses $\alpha \neq \beta$. If α, β first differ at position k (both defined and unequal there), the clopen sets $\mathcal{H}_\partial[\alpha|_k]$ and its complement separate the two fibers, so C cannot meet both. Otherwise one address is a finite word u and the other strictly extends it; then β extends ua for some $a \in \mathtt{A}$, the fiber of β lies in the clopen set $\mathcal{H}_\partial[ua]$, and K_u lies in its complement by Lemma 5.1(c). In both cases C is disconnected by a clopen partition, a contradiction. So the components of $\mathcal{H}_\partial$ are exactly the fibers.

(v). From $g \circ \phi_a = \mathrm{id}_P$ we get $g \circ \phi_{aw} = \phi_w$, hence $g(K_{aw}) = K_w$. Since $g\colon Q_a \to P$ is a homeomorphism (H2), it commutes with the nested intersection (5.2):

$$g(E_{a\omega}) = g\Big(\bigcap_m Q_{(a\omega)|m}\Big) = \bigcap_m g\big(Q_{a(\omega|m-1)}\big) = \bigcap_m Q_{\omega|m-1} = E_\omega.$$

Each transition is the restriction of f^q to a subset of Q_a, and the deleted first symbol a records the inverse sheet. □

5.3. **The component quotient and the exact partial shift.**

Theorem 5.4 (Component quotient)**.** *Define $\pi_\partial\colon \mathcal{H}_\partial \to \Omega_\partial$ by $\pi_\partial(x) = \alpha \iff x \in \mathrm{fib}(\alpha)$. Then π_∂ is a well-defined continuous closed surjection whose fibers are precisely the connected components of $\mathcal{H}_\partial$. Consequently the component space of $\mathcal{H}_\partial$, with the quotient topology, is homeomorphic to Ω_∂:*

$$\mathcal{H}_\partial/\sim_{component} \cong\ \Omega_\partial.$$

In particular the component decomposition of $\mathcal{H}_\partial$ is upper semicontinuous.

Proof. Well-definedness and surjectivity hold because the nonempty fibers are pairwise disjoint and cover $\mathcal{H}_\partial$ (Lemma 5.1(d), Theorem 5.3(ii)). For continuity, let $x_j \to x$ in $\mathcal{H}_\partial$ and write $\alpha_j = \pi_\partial(x_j)$, $\alpha = \pi_\partial(x)$. If $\alpha_j \not\to \alpha$, compactness of Ω_∂ gives a subsequence $\alpha_{j_k} \to \beta \neq \alpha$; Lemma 5.2 then puts $x \in \mathrm{fib}(\beta)$, while $x \in \mathrm{fib}(\alpha)$, contradicting disjointness of fibers. Hence $\alpha_j \to \alpha$ and π_∂ is continuous. A continuous map from a compact space to a Hausdorff space is closed; a closed continuous surjection is a quotient map, and its fibers are the components by Theorem 5.3(iii). Upper semicontinuity of the decomposition is exactly closedness of the quotient map with compact fibers. □

Theorem 5.5 (Exact partial shift realization)**.** *For $a \in \mathtt{A}$ put $H_a = \mathcal{H}_\partial \cap Q_a$. Then:*

(i) *$H_a = \phi_a(\mathcal{H}_\partial)$, and H_0, H_1 are disjoint clopen subsets of $\mathcal{H}_\partial$ with*

$$\mathcal{H}_\partial \setminus K_\epsilon = H_0 \sqcup H_1;$$

(ii) *$g|_{H_a}\colon H_a \to \mathcal{H}_\partial$ is a homeomorphism with inverse ϕ_a; hence*

$$g\colon \mathcal{H}_\partial \setminus K_\epsilon \longrightarrow \mathcal{H}_\partial$$

is a (trivial) two-sheeted covering with clopen sheets H_0, H_1;

(iii) *the component quotient semiconjugates this covering to the first-symbol deletion shift:*

$$\pi_\partial \circ g = \sigma_{\mathrm{pre}} \circ \pi_\partial \qquad \text{on } \mathcal{H}_\partial \setminus K_\epsilon.$$

Proof. (i) The identities $\phi_a(K_w) = K_{aw}$ and $\phi_a(E_\omega) = E_{a\omega}$ show $\phi_a(\mathcal{H}_\partial) = \bigcup_w K_{aw} \cup \bigcup_\omega E_{a\omega} = \mathcal{H}_\partial[a]$, which equals $\mathcal{H}_\partial \cap Q_a$ and is clopen by the proof of Theorem 5.3(iii). Every fiber other than the root fiber has an address beginning with 0 or 1, so $\mathcal{H}_\partial \setminus K_\epsilon = H_0 \sqcup H_1$, and $H_0 \cap H_1 = \varnothing$ by (H1).

(ii) On $H_a \subset Q_a$ the map g is the restriction of the homeomorphism $g\colon Q_a \to P$ of (H2), with inverse ϕ_a; by (i), ϕ_a carries $\mathcal{H}_\partial$ onto H_a, so $g|_{H_a}$ is a homeomorphism onto $\mathcal{H}_\partial$. The two sheets are clopen and disjoint, so the restriction of g to their union is a trivial two-sheeted covering of $\mathcal{H}_\partial$.

(iii) On H_a, the fiber of address $a\beta$ maps onto the fiber of address β by (5.5); this is exactly the semiconjugacy identity. □

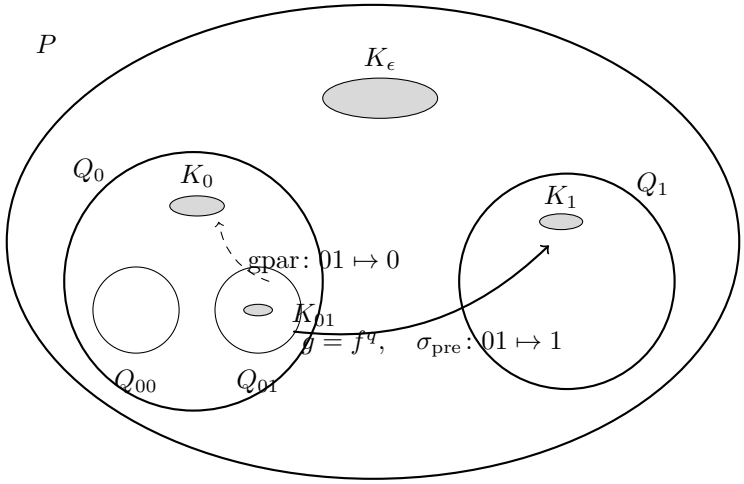


FIGURE 1. The two parent maps at address 01. Geometrically $Q_{01} = \phi_0(Q_1) \subset Q_0$, so deleting the *last* symbol moves to the surrounding cylinder and its fiber, $K_{01} \rightsquigarrow K_0$ (dashed arrow; containment, no dynamics). Dynamically $g(Q_{01}) = Q_1$ and $g(K_{01}) = K_1$, so deleting the *first* symbol is the actual transition by f^q (solid arrow). The root K_ϵ sits in the first-level gap $P \setminus (Q_0 \cup Q_1)$; the fiber $K_0 = \phi_0(K_\epsilon)$ sits in the gap of Q_0 between its children Q_{00}, Q_{01}.

5.4. **The two parent maps and the root arrow.**

Remark 5.6 (Two parent maps). The package carries two structurally different parent arrows (Figure 1):

$$\mathrm{gpar}(wa) = w, \qquad \sigma_{\mathrm{pre}}(a\alpha) = \alpha. \tag{5.6}$$

The *geometric parent* gpar deletes the *last* symbol of a nonempty finite word. It is realized by cylinder inclusion, $Q_{wa} \subset Q_w$, and by *no* dynamical map: there is in general no iterate of f carrying K_{wa} onto K_w. It has no action on ends, which have no last symbol.

The *dynamical parent* σ_{pre} deletes the *first* symbol. It is defined on all non-root finite addresses and on *all* ends, and it is realized by the exact two-branch covering of Theorem 5.5, with transition time q and the deleted symbol as inverse sheet. Concretely, at address 01: $\mathrm{gpar}(01) = 0$ because $Q_{01} = \phi_0(Q_1) \subset Q_0$, while $\sigma_{\mathrm{pre}}(01) = 1$ because $g(K_{01}) = K_1$. The two arrows agree on no address of length ≥ 2 in general, and conflating them collapses the distinction between nesting depth and dynamical time. Both are retained as separate typed data.

Remark 5.7 (The root arrow is separate data). The dynamical parent σ_{pre} is undefined at the empty word. The root has its own transition, coming from the terminal cycle of Proposition 2.1:

$$\tau_{\mathrm{root}}: \quad g(K_\epsilon) = f^q(K_\epsilon) = \mathcal{K}_{q \bmod p}. \tag{5.7}$$

This arrow lands on a component of the terminal cycle; its target is indexed by the cycle rather than by an address, and no positional claim about $\mathcal{K}_{q \bmod p}$ relative to $\mathcal{H}_\partial$ is made beyond Proposition 2.1. It is *not* an extension of σ_{pre}, and no assertion is made that the image escapes $\mathcal{H}_\partial$ permanently (Non-claims 2.3(N1)). Keeping τ_{root} as a separate typed arrow, rather than force-fitting it into the shift, is precisely what makes the package close up without a circular hypothesis.

5.5. **Naturality under root enlargement.** The virtual program enlarges its root package. The compactification is functorial under exactly the enlargements that stay in the gap. For a compact connected R with $K_\epsilon \subset R \subset P \setminus (Q_0 \cup Q_1)$, write

$$\mathcal{H}_\partial(R) = \bigcup_{w \in \mathtt{A}^{<\mathbb{N}}} \phi_w(R) \;\cup\; \bigcup_{\omega \in \mathtt{A}^{\mathbb{N}}} E_\omega$$

for the compactum with root R; thus $\mathcal{H}_\partial = \mathcal{H}_\partial(K_\epsilon)$.

Theorem 5.8 (Root-gap enlargement naturality)**.** *Let*

$$K_\epsilon \subset S \subset S^+ \subset P \setminus (Q_0 \cup Q_1) \tag{5.8}$$

be compact connected roots. Then for each $R \in \{S, S^+\}$:

(i) *all separations of Lemma 5.1, the conclusion of Lemma 5.2, and conclusions (i), (iii), (iv), (v) of Theorem 5.3 hold for $\mathcal{H}_\partial(R)$; $\mathcal{H}_\partial(R)$ is compact and every fiber is a nonempty continuum;*

(ii) *the component quotient and partial-shift theorems (Theorems 5.4 and 5.5) hold for $\mathcal{H}_\partial(R)$, with root fiber R in place of K_ϵ;*

(iii) *membership in the filled Julia set is exactly*

$$\mathcal{H}_\partial(R) \subset K(f) \iff R \subset K(f),$$

and under (H3) the same equivalence holds with $J(f)$;

(iv) *the inclusions $\phi_w(S) \subset \phi_w(S^+)$ commute with g at every finite word, and the end fibers are unchanged.*

Proof. (i)–(ii) The proofs of the listed conclusions use the root only through compactness, connectedness, containment in P, and disjointness from $Q_0 \cup Q_1$—the consequences of (2.5) actually invoked there—all of which hold for S and S^+ by (5.8). The only step of Theorem 5.3(ii) not covered is the containment of the fibers in $K(f)$, which used $K_\epsilon \subset K(f)$ and is treated separately in (iii). The quotient and partial-shift proofs use only the separations, the label-limit lemma, compactness, and (H1)–(H2), so they persist verbatim.

(iii) If $R \subset K(f)$, complete invariance of $K(f)$ gives $\phi_w(R) \subset f^{-q|w|}(K(f)) = K(f)$, and the end fibers lie in $K(f)$ by the bounded-orbit argument of Theorem 5.3(ii); hence $\mathcal{H}_\partial(R) \subset K(f)$. Conversely R is the root fiber of $\mathcal{H}_\partial(R)$, so $\mathcal{H}_\partial(R) \subset K(f)$ forces $R \subset K(f)$. Under (H3), $K(f) = J(f)$.

(iv) Injectivity of ϕ_w gives $\phi_w(S) \subset \phi_w(S^+)$, and $g \circ \phi_a = \mathrm{id}$ gives the commuting squares. The end fibers (5.2) depend only on P, Q_0, Q_1. □

Remark 5.9 (Three levels of enlargement, and the root arrow). Theorem 5.8 is the *topological* level: it needs only a compact connected root in the gap, and it does not assert that S or S^+ is full, contained in $J(f)$, dynamically protected, access-complete, or admissible as an analytic virtual carrier; those properties are separate audits. The *fullness/cell* level requires a full root compactly contained in the open gap component G_0 and is treated in Corollary 6.7. The *Julia* level is the equivalence (iii). Finally, the root arrow (5.7) refers to the terminal cycle of the original root K_ϵ; for an enlarged root R the image $g(R)$ carries no terminal-cycle identification, and any dynamical meaning of the enlarged root arrow requires a separate audit.

6. Fullness, the end compactum, gap cells, and replacement

6.1. **Full fibers and connected complement.**

Theorem 6.1 (Full fibers and nonseparation)**.** *Every fiber K_w and E_ω of Theorem 5.3 is a full continuum, and $\widehat{\mathbb{C}} \setminus \mathcal{H}_\partial$ is connected. Moreover, for every Jordan domain U with $P \Subset U$, the relative complement $U \setminus \mathcal{H}_\partial$ is connected.*

Proof. The root K_ϵ is full [DH85]. Each $\phi_w \colon P \to Q_w$ is an orientation-preserving homeomorphism of closed Jordan disks, so $K_w = \phi_w(K_\epsilon)$ is full by Lemma 3.3(a). For an end,

$$\widehat{\mathbb{C}} \setminus E_\omega = \bigcup_{m \geq 1} \left(\widehat{\mathbb{C}} \setminus Q_{\omega|m}\right)$$

is an increasing union of connected sets containing ∞, hence connected; so E_ω is full.

For $m \geq 0$ define the approximants

$$\mathcal{H}_\partial^{(m)} = \bigcup_{|u|<m} K_u \;\cup\; \bigcup_{|w|=m} Q_w. \tag{6.1}$$

Each term is full, and the terms of $\mathcal{H}_\partial^{(m)}$ are pairwise disjoint: equal-depth cylinders are disjoint (Lemma 5.1(b)), distinct finite fibers are disjoint ((d)), and for $|u| < m = |w|$ the word u is either incomparable to w (disjoint cylinders) or a strict prefix of w (then $K_u \cap Q_w = \varnothing$ by (c)). The sets (6.1) decrease in m: each K_u with $|u| = m$ lies in Q_u, and each Q_w with $|w| = m+1$ lies in $Q_{w|m}$.

Their intersection is exactly $\mathcal{H}_\partial$. Indeed $\mathcal{H}_\partial \subset \mathcal{H}_\partial^{(m)}$ for every m (a fiber with address of length $\geq m$ or an end fiber lies in its depth-m cylinder). Conversely, let $x \in \bigcap_m \mathcal{H}_\partial^{(m)}$. If x lies in some finite fiber K_u we are done. Otherwise, for every m there is a unique word w_m of length m with $x \in Q_{w_m}$ (uniqueness by disjointness of equal-depth cylinders), and $x \in Q_{w_{m+1}} \subset Q_{w_{m+1}|m}$ forces $w_{m+1}|m = w_m$. The words w_m therefore converge to an end ω with $x \in \bigcap_m Q_{\omega|m} = E_\omega$.

Lemma 3.2 applied to the decreasing packets (6.1) proves that $\widehat{\mathbb{C}} \setminus \mathcal{H}_\partial$ is connected; its localized clause, applied with the exterior full continuum $\widehat{\mathbb{C}} \setminus U$ adjoined to each packet, proves that $U \setminus \mathcal{H}_\partial$ is connected. □

6.2. **The end compactum.**

Corollary 6.2 (Cantor-indexed end compactum)**.** *Let*

$$\mathcal{H}_\infty = \bigcup_{\omega \in \mathtt{A}^{\mathbb{N}}} E_\omega.$$

Then

$$\mathcal{H}_\infty = \bigcap_{m \geq 1} \bigcup_{|w|=m} Q_w, \tag{6.2}$$

$\mathcal{H}_\infty$ is compact, its connected components are precisely the end fibers E_ω, and $\widehat{\mathbb{C}} \setminus \mathcal{H}_\infty$ is connected. The component quotient restricts to a continuous closed surjection $\mathcal{H}_\infty \to \mathtt{A}^{\mathbb{N}} = \{0,1\}^{\mathbb{N}}$, so the component space of $\mathcal{H}_\infty$ is the Cantor space, and

$$g \colon \mathcal{H}_\infty \longrightarrow \mathcal{H}_\infty$$

is a trivial two-sheeted covering of $\mathcal{H}_\infty$ over itself, semiconjugate under the quotient to the one-sided binary shift, realized by the two univalent branches ϕ_0, ϕ_1 of f^{-q}.

Proof. For (6.2): every E_ω lies in $Q_{\omega|m}$ for all m, giving one inclusion. Conversely, a point x of the right-hand side lies, for every m, in a unique depth-m cylinder Q_{w_m}, and uniqueness forces $w_{m+1}|m = w_m$ as in Theorem 6.1; hence $x \in E_\omega$ for $\omega = \lim w_m$. No finite fiber meets

the right-hand side: for $m > |u|$, K_u misses every depth-m cylinder (Lemma 5.1(c) and (b)). Thus (6.2) holds, and $\mathcal{H}_\infty$ is an intersection of compacta, hence compact.

The levels $\bigcup_{|w|=m} Q_w$ are finite unions of pairwise disjoint full continua and decrease in m, so Lemma 3.2 makes $\widehat{\mathbb{C}} \setminus \mathcal{H}_\infty$ connected. Components: the sets $\mathcal{H}_\infty \cap Q_u$ are clopen in $\mathcal{H}_\infty$ (they are the restrictions of the clopen sets $\mathcal{H}_\partial[u]$ of Theorem 5.3), and the argument of Theorem 5.3(iii) applies verbatim with only end addresses present. The quotient statement restricts Theorem 5.4; $\mathtt{A}^{\mathbb{N}} \subset \Omega_\partial$ is closed, and the restricted map is again a continuous closed surjection. Finally $\phi_a(\mathcal{H}_\infty) = \mathcal{H}_\infty \cap Q_a$ and $g(E_{a\omega}) = E_\omega$ give the two-sheeted covering of $\mathcal{H}_\infty$ over itself and the semiconjugacy to the full one-sided shift, exactly as in Theorem 5.5; here no root is removed, since σ_{pre} is total on ends. □

Remark 6.3. $\mathcal{H}_\infty$ is the statement most directly comparable to a "Cantor family of continua": a compact nonseparating union of full continua whose component space is $\{0,1\}^{\mathbb{N}}$ and on which the binary shift is realized by two actual univalent inverse branches. It does not identify the E_ω as small Julia sets (Non-claims 2.3(N2)); it realizes exactly the topological and dynamical skeleton.

6.3. **The gap-cell atlas.** Recall G_0, the component of $\operatorname{int} P \setminus (Q_0 \cup Q_1)$ containing K_ϵ (Standing Hypotheses 2.2(H4)). By Lemma 3.4 choose once and for all a closed Jordan disk D with

$$K_\epsilon \subset \operatorname{int} D \subset D \Subset G_0, \tag{6.3}$$

and put $D_w = \phi_w(D)$.

Theorem 6.4 (Gap-cell atlas)**.** *The closed Jordan cells D_w are pairwise disjoint—including for ancestor–descendant pairs—and satisfy*

$$K_w \subset \operatorname{int} D_w, \qquad D_w \Subset \operatorname{int} Q_w \setminus (Q_{w0} \cup Q_{w1}), \tag{6.4}$$

$$D_w \cap \mathcal{H}_\partial = K_w, \qquad g(D_{aw}) = D_w. \tag{6.5}$$

If $S_w \Subset \operatorname{int} D_w$ is compact and connected, then

$$\operatorname{Fill}(S_w) \Subset \operatorname{int} D_w. \tag{6.6}$$

Proof. By (6.3) and (H2), ϕ_w carries $\operatorname{int} P$ to $\operatorname{int} Q_w$, Q_a to Q_{wa}, and D to D_w, which proves (6.4). For disjointness: incomparable words have disjoint cylinders containing their cells; and if $v = uat$ is a strict descendant of u, then D_v lies in the child cylinder Q_{ua} while D_u misses both children of Q_u by (6.4).

First identity of (6.5): fibers with address strictly extending w lie in the children $Q_{w0} \cup Q_{w1}$, which D_w misses; fibers with address incomparable to w miss $Q_w \supset D_w$; fibers K_u with u a strict prefix of w miss $Q_{w|_{|u|+1}} \supset D_w$ by Lemma 5.1(c); end fibers extending w lie in a child of Q_w. What remains is $K_w \subset \operatorname{int} D_w$. The second identity follows from $g \circ \phi_a = \mathrm{id}$.

For (6.6): a collar of ∂D_w together with the exterior of D_w is a connected subset of the unbounded component of $\mathbb{C} \setminus S_w$; every bounded complementary component of S_w is therefore contained away from that collar, inside $\operatorname{int} D_w$. □

6.4. **Cellwise replacement at arbitrary addresses.** The next theorem removes the finiteness restriction of the packet replacement used in Draft I. Replacement may occur at *every* finite address simultaneously; finiteness is required only where the connector construction of Section 7 actually uses it.

Theorem 6.5 (Cellwise packet replacement at arbitrary addresses)**.** *Let $F \subset \mathtt{A}^{<\mathbb{N}}$ be an arbitrary—possibly infinite—set of finite addresses, and for each $w \in F$ let L_w be a compact*

connected set with

$$K_w \subset L_w \Subset \operatorname{int} D_w. \tag{6.7}$$

Put $\widetilde{K}_w = L_w$ *for* $w \in F$ *and* $\widetilde{K}_w = K_w$ *otherwise, and*

$$\widetilde{\mathcal{H}} = \bigcup_{w \in \mathtt{A}^{<\mathbb{N}}} \widetilde{K}_w \;\cup\; \bigcup_{\omega \in \mathtt{A}^{\mathbb{N}}} E_\omega. \tag{6.8}$$

Then:

(i) $\widetilde{\mathcal{H}}$ *is compact; its connected components are exactly the sets* $\widetilde{K}_w$ *(*$w \in \mathtt{A}^{<\mathbb{N}}$*) and* E_ω *(*$\omega \in \mathtt{A}^{\mathbb{N}}$*); the realization* $\alpha \mapsto \widetilde{\mathrm{fib}}(\alpha)$ *is upper semicontinuous; and the component projection* $\widetilde{\pi}\colon \widetilde{\mathcal{H}} \to \Omega_\partial$ *is a continuous closed surjection.*

(ii) *If in addition every* L_w *(*$w \in F$*) is full, then* $\widehat{\mathbb{C}} \setminus \widetilde{\mathcal{H}}$ *is connected, and for every Jordan domain* U *with* $P \Subset U$ *the relative complement* $U \setminus \widetilde{\mathcal{H}}$ *is connected.*

The containment $K_w \subset L_w$ *is used nowhere in the proof of* (i)*; it records the enlargement semantics and may be dropped.*

Proof. Define the modified approximants

$$\widetilde{\mathcal{H}}^{(m)} = \bigcup_{|u|<m} \widetilde{K}_u \;\cup\; \bigcup_{|w|=m} Q_w. \tag{6.9}$$

At each fixed m this is a *finite* union—fewer than 2^m packets and exactly 2^m cylinders—regardless of the cardinality of F. The terms are pairwise disjoint: distinct packets lie in the pairwise disjoint cells D_u (both K_u and L_u lie in $\operatorname{int} D_u$ by (6.4) and (6.7)); a packet $\widetilde{K}_u$ with $|u| < m$ and a cylinder Q_w with $|w| = m$ are disjoint because either u is a strict prefix of w, in which case Q_w lies in a child of Q_u, which $D_u \supset \widetilde{K}_u$ misses by (6.4), or u and w are incomparable, in which case already $Q_u \cap Q_w = \varnothing$. The sets (6.9) decrease in m: each $\widetilde{K}_u$ with $|u| = m$ lies in $\operatorname{int} D_u \subset Q_u$, and each depth-$(m{+}1)$ cylinder lies in its parent.

The intersection of the modified approximants is exactly $\widetilde{\mathcal{H}}$: one inclusion is clear as before, and if $x \in \bigcap_m \widetilde{\mathcal{H}}^{(m)}$ lies in no packet $\widetilde{K}_u$, then at every depth it lies in a unique cylinder, the words nest, and x lies in the corresponding end fiber. Compactness of $\widetilde{\mathcal{H}}$ follows, each approximant being compact.

For the component structure, upper semicontinuity, and the quotient in (i), the arguments of Theorems 5.3 and 5.4 apply verbatim once their two inputs are re-verified for the modified fibers. First, the separations: distinct packets lie in disjoint cells; a packet $\widetilde{K}_u$ misses every cylinder Q_v with v strictly extending u (it lies in D_u, which misses the children of Q_u) and every cylinder incomparable to u; hence distinct modified fibers are pairwise disjoint, and the clopen sets $\widetilde{\mathcal{H}} \cap Q_u$ again consist exactly of the modified fibers with address extending u. Second, the label-limit lemma: if $\alpha_j \to \alpha$ and $x_j \in \widetilde{\mathrm{fib}}(\alpha_j) \to x$, then for finite α the labels are eventually constant and $\widetilde{K}_\alpha$ is closed, while for an end α every modified fiber with address extending $\alpha|m$ lies in $Q_{\alpha|m}$ (packets via $\widetilde{K}_v \subset D_v \subset Q_v \subset Q_{\alpha|m}$), so $x \in E_\alpha$. With these two inputs, the clopen-partition components argument, the closed-graph upper semicontinuity, and the continuity-closedness of the component projection repeat word for word.

For (ii), every term of (6.9) is now full (L_w by hypothesis, K_u and Q_w as before), so Lemma 3.2 and its localized clause apply to the decreasing packets, with the exterior continuum $\widehat{\mathbb{C}} \setminus U$ adjoined for the relative statement. □

Remark 6.6. No uniform diameter estimate, no coherence between the L_w at different addresses, and no dynamical compatibility is required in Theorem 6.5; correspondingly, no equivariance is

asserted for the replaced package—$g(L_{aw})$ need not equal L_w. Equivariance returns exactly for coherent families $L_w = \phi_w(S)$, which is the content of Corollary 6.7. For nonfull replacements, part (i) survives while part (ii) genuinely fails in general (Appendix B, Test B.6).

6.5. **Full root-gap naturality.**

Corollary 6.7 (Full root-gap naturality)**.** *Let S be a nondegenerate full continuum with*

$$K_\epsilon \subset S \Subset G_0,$$

and put $S_w = \phi_w(S)$. Then the full package holds for the root S:

(i) *$\mathcal{H}_\partial(S) = \bigcup_w S_w \cup \bigcup_\omega E_\omega$ is compact, its components are exactly the S_w and E_ω, the component projection onto Ω_∂ is a continuous closed surjection, and $g\colon \mathcal{H}_\partial(S) \setminus S \to \mathcal{H}_\partial(S)$ is a trivial two-sheeted covering with $g(S_{aw}) = S_w$;*

(ii) *every S_w is full, $\widehat{\mathbb{C}} \setminus \mathcal{H}_\partial(S)$ is connected, and $U \setminus \mathcal{H}_\partial(S)$ is connected for every Jordan domain U with $P \Subset U$;*

(iii) *there is a gap-cell atlas for S: a closed Jordan disk D_S with $S \subset \operatorname{int} D_S \subset D_S \Subset G_0$ whose transported cells $D_{S,w} = \phi_w(D_S)$ satisfy all conclusions of Theorem 6.4 with S_w in place of K_w;*

(iv) *$\mathcal{H}_\partial(S) \subset K(f)$ if and only if $S \subset K(f)$; under (H3) the same holds with $J(f)$.*

Proof. The root S is compact, connected, nondegenerate, full, and satisfies the gap relation $S \subset G_0 \subset \operatorname{int} P \setminus (Q_0 \cup Q_1)$; that is, S satisfies (H4) verbatim. Part (i) is Theorem 5.8(i)–(ii) with $R = S$, together with $g \circ \phi_{aw} = \phi_w$ for the displayed equivariance. Part (ii) reruns the proof of Theorem 6.1 with root S: that proof uses only fullness of the root, the separations, and (H2), all available. Part (iii) reruns Theorem 6.4: Lemma 3.4 applies to the nondegenerate full continuum $S \Subset G_0$ and produces D_S, and the proof of Theorem 6.4 uses only (6.3)-type containment, (H2), and the separations. Part (iv) is Theorem 5.8(iii). □

7. The finite protected carrier

The final theorem combines the package with the connector technology of Section 4. It is the dynamical payoff: finitely many protected accesses at the root transport to finitely many selected fibers, and a connected full carrier realizes them simultaneously while meeting the compactum in exactly the selected fibers. Here finiteness is genuinely used—the connector is a finite tree—in contrast to the replacement theorem, which is not so restricted.

Lemma 7.1 (Proper protected impressions leave a port)**.** *Let $K \Subset \operatorname{int} D$ be a nondegenerate full continuum in a closed Jordan disk D and let $P_{\mathrm{prot}} \subsetneq \partial K$ be compact, possibly empty. There are $x \in \partial K \setminus P_{\mathrm{prot}}$ and a Jordan arc α from x to ∂D, otherwise in $\operatorname{int} D \setminus K$. Moreover, finitely many crosscut chains of $\widehat{\mathbb{C}} \setminus K$ whose impressions have union contained in P_{prot} can be truncated so that, writing S for the connected union of K with the truncated relative collar tails,*

$$\alpha \cap \operatorname{Fill}(S) = \{x\}. \tag{7.1}$$

Proof. If $P_{\mathrm{prot}} = \varnothing$, choose any $y \in \partial K$ and skip the distance argument below (there are then no chains to truncate, $S = K$, and only the port construction is needed). Otherwise choose $y \in \partial K \setminus P_{\mathrm{prot}}$ and put $\delta = \operatorname{dist}(y, P_{\mathrm{prot}}) > 0$. Choose $z \in \operatorname{int} D \setminus K$ with $|z - y| < \delta/4$, and let $x \in K$ minimize distance to z. The segment $(x, z]$ misses K, and $|x - y| \le 2|z - y| < \delta/2$, so $x \notin P_{\mathrm{prot}}$. The complement of the full continuum K in $\widehat{\mathbb{C}}$ is an open connected, hence arcwise connected, set; join z to the exterior of D, take a simple subarc of the concatenation starting at x, and stop at its first hit on ∂D. This is α.

The compact arc α misses the compact set P_{prot} at positive distance. For each chain, the nested compact sets $\overline{D_n}$ have intersection equal to the impression, which lies in P_{prot}; hence sufficiently deep relative tails lie in prescribed neighborhoods of the impressions chosen to miss α. Truncate each chain there. Finally, extending α beyond ∂D to ∞ shows that every point of α except x lies in the unbounded component of $\mathbb{C} \setminus S$, which proves (7.1). □

Theorem 7.2 (Finite direct-decoration protected carrier)**.** *Fix a nonempty finite set $F \subset \mathtt{A}^{<\mathbb{N}}$ of addresses and finitely many (possibly zero) crosscut chains of $\widehat{\mathbb{C}} \setminus K_\epsilon$ (the* protected root chains*) whose relative terminal collar tails are pairwise disjoint and whose total impression is a proper compact subset $P_{\mathrm{prot}} \subsetneq \partial K_\epsilon$. Let U be a Jordan domain with $P \Subset U$. Then the chains can be truncated and transported by the inverse branches ϕ_w ($w \in F$), and there exist a free port point $x_w \in \partial K_w$ and a port arc α_w for each $w \in F$, such that:*

(i) *the transported packet fullifications $L_w = \mathrm{Fill}\big(K_w \cup (\text{transported relative collar tails})\big)$ satisfy $K_w \subset L_w \Subset \operatorname{int} D_w$, so they lie in the pairwise disjoint gap cells of Theorem 6.4;*

(ii) *there is a compact connected full carrier $H \Subset U$ containing every selected fiber K_w ($w \in F$) with singleton attachment trace $\overline{H \setminus K_w} \cap K_w = \{x_w\}$, and meeting the compactum in exactly the selected fibers:*

$$H \cap \mathcal{H}_\partial = \bigcup_{w \in F} K_w; \tag{7.2}$$

(iii) *every transported protected chain survives as a prime-end chain of $\widehat{\mathbb{C}} \setminus H$, and the survival map is injective and preserves the cyclic order of the transported family in $\widehat{\mathbb{C}} \setminus H$;*

(iv) *no local connectivity of any fiber is assumed or concluded.*

Proof. Step 1: root port and truncation. Apply Lemma 7.1 with $K = K_\epsilon$, the cell D of (6.3), and P_{prot} (the lemma covers $P_{\mathrm{prot}} = \varnothing$): this yields a free point $x \in \partial K_\epsilon \setminus P_{\mathrm{prot}}$, an arc α from x to ∂D otherwise in $\operatorname{int} D \setminus K_\epsilon$, and truncations of the chains such that, for the connected union S of K_ϵ with the truncated relative collar tails, (7.1) holds. Truncating further if necessary, $S \Subset \operatorname{int} D$: the sets $\overline{D_n}$ shrink to impressions on $\partial K_\epsilon \subset \operatorname{int} D$. The truncated relative tails are pairwise disjoint: by the null-chain condition (3.4), each truncated relative tail lies in the open terminal collar of its own chain, and those collars contain the pairwise disjoint original relative terminal tails of the hypothesis—so distinct chains' truncated tails lie in disjoint sets.

Step 2: transport. For $w \in F$ put

$$S_w = \phi_w(S), \qquad L_w = \mathrm{Fill}(S_w), \qquad x_w = \phi_w(x), \qquad \alpha_w = \phi_w(\alpha).$$

Each ϕ_w restricts to an orientation-preserving homeomorphism of closed Jordan disks $D \to D_w$ ((H2), Theorem 6.4). By the pointed Schoenflies transport (Lemma 3.3(b)), it commutes with Fill on compact subsets of D, carries crosscut chains of $\widehat{\mathbb{C}} \setminus K_\epsilon$ to crosscut chains of $\widehat{\mathbb{C}} \setminus K_w$ preserving selected sides, the null-chain condition, and cyclic order, and transports (7.1) to

$$\alpha_w \cap L_w = \{x_w\}. \tag{7.3}$$

Since $S \Subset \operatorname{int} D$ is compact and connected, Theorem 6.4(6.6) gives $L_w = \mathrm{Fill}(S_w) \Subset \operatorname{int} D_w$, proving (i). The arc α_w runs from x_w to ∂D_w, otherwise in $\operatorname{int} D_w \setminus L_w$.

Step 3: one free component. The data (6.7) hold for the finite family $\{L_w\}_{w \in F}$ with every L_w full, so Theorem 6.5(ii) applies: $U \setminus \widetilde{\mathcal{H}}$ is connected, where $\widetilde{\mathcal{H}}$ is the replaced compactum (6.8). Set the connector inventory of Section 4 as follows: atoms $A_w := K_w$ ($w \in F$); protected collars: the transported truncated relative collar tails, which are pairwise disjoint—within one fiber by Step 1 and injectivity of ϕ_w, across fibers because they lie in the pairwise disjoint

cells D_w; forbidden set $Z := \mathcal{H}_\partial \setminus \bigcup_{w\in F} K_w$ (every unselected finite fiber and every end fiber), which is compact because the selected fibers are clopen in $\mathcal{H}_\partial$. Then the frozen set Q of (4.3) satisfies $Q \subset \widetilde{\mathcal{H}}$: atoms and relative collar tails lie in the packets L_w, and $Z \subset \widetilde{\mathcal{H}}$. Hence

$$U \setminus \widetilde{\mathcal{H}} \;\subset\; \Omega_{\mathrm{free}} = U \setminus Q,$$

and the connected set $U \setminus \widetilde{\mathcal{H}}$ lies in one component Ω_0 of Ω_{free}.

The port interiors lie in that component: by (7.3) and (6.5),

$$\alpha_w((0,1]) \;\subset\; \big(\operatorname{int} D_w \setminus L_w\big) \cup \partial D_w \;\subset\; U \setminus \widetilde{\mathcal{H}},$$

because $D_w \cap \widetilde{\mathcal{H}} = L_w$ (the cell meets the replaced compactum exactly in its own packet, by (6.5) and disjointness of the cells) and ∂D_w misses $\widetilde{\mathcal{H}}$. Also $\alpha_w \cap K_w \subset \alpha_w \cap L_w = \{x_w\}$ with $x_w \in \partial K_w$ free: $x = \phi_w^{-1}(x_w)$ is not in P_{prot} and hence x_w is not an impression of a transported chain. The arcs α_w are pairwise disjoint, lying in the pairwise disjoint closed cells D_w. Thus Ω_0 has simultaneous singleton ports to every atom.

Step 4: certificate and carrier. Theorem 4.3 yields a protected ribbon-routing certificate for the atoms K_w, protected collars, and forbidden set Z, with connector tree $T_{\mathfrak{R}}$ satisfying (4.1), and Theorem 4.2 realizes it: a compact connected full carrier $H = H_{\mathfrak{R}} \Subset U$ containing every K_w with singleton attachment traces $\{x_w\}$, missing Z, with every transported chain surviving as a prime-end chain of $\widehat{\mathbb{C}} \setminus H$, injectively and in the cyclic order of the transported family. For the exact trace (7.2): $H \supset \bigcup_{w\in F} K_w$ by construction, and $H \setminus \bigcup_{w\in F} K_w = T_{\mathfrak{R}} \setminus \{x_w : w \in F\}$ lies in $\Omega_{\mathrm{free}} \cup \{\alpha_w(1)\}$, which misses $Z \cup \bigcup_{w\in F} K_w = \mathcal{H}_\partial$ except at the port points $x_w \in K_w$ already counted; hence $H \cap \mathcal{H}_\partial$ is exactly the union of the selected fibers. This proves (ii) and (iii). The arguments use crosscut chains and plane topology only, never a boundary parametrization of any fiber, proving (iv). □

Remark 7.3 (Finite and direct-scale only). Theorem 7.2 is finite and direct-scale only. It does not construct a second satellite portrait, an end-fiber return map, a uniform incidence bound, or the future fjord, dam, and attachment data of the virtual program (Non-claims 2.3). Selecting *infinitely* many fibers, or an end fiber, as atoms is not covered by the theorem—the connector is a finite tree—although the underlying packet compactum need not be finite (Theorem 6.5). Proposition 8.1 shows that the raw marking data grow without bound as the decoration depth increases.

8. Scope and open problems

8.1. **Why the remaining finiteness restriction is honest.**

Proposition 8.1 (Raw decoration-fan growth)**.** *The quadratic separator $Y^{(n-1)q+1}$ has exactly 2^n vertices, and the decoration construction supplies a distinct pairwise-disjoint univalent pullback Q^v for every vertex* [KL08]*. Hence any marking that retains one distinct record for every source Q^v has at least 2^n entries and admits no universal cardinality bound as n varies.*

Proof. The vertex count and the one-per-vertex pairwise disjoint pullbacks are the source statements in [KL08]. Distinct disjoint pullbacks require distinct marked records, so the cardinality is at least 2^n, which is unbounded in n. □

Thus a virtual construction over all scales requires either a uniformly bounded proof-complete subset of the raw fan or a future-faithful access compression; neither is constructed here (Non-claims 2.3(N4)). Note the division of labor after Theorem 6.5: the packet *compactum* supports arbitrary, even infinite, cellwise modification, while the protected *connector* remains finite.

8.2. **What separates this model from the virtual package.** Dudko's virtual Julia sets are described as connected hulls of Cantor families of small Julia sets carrying part of the critical orbit [Dud25]. The present package supplies the correctly typed skeleton of such an object at one scale: address space, embedded components, end fibers, the component quotient, both parent maps, root arrow, cells, arbitrary cellwise replacement, and protected carrier. What it does not supply is exactly what remains open:

(a) *Identification of ends.* The end fibers E_ω are nested puzzle intersections. They have not been identified with small Julia sets, and no renormalization structure on them is claimed ((N2)).
(b) *Analytic end dynamics.* The covering of Theorem 5.5 moves ends by deleting the first symbol; no first-return or self-return map *on* a single end fiber is constructed ((N3)).
(c) *Portraits and markings.* No second satellite portrait is constructed, and the transported prime-end data of Theorem 7.2 concern finitely many chains at finitely many finite fibers only ((N2), (N4)).

The two open problems below record the missing constructions in the typing of this paper. They are stated for the checked sources [Dud25, KKL26]; solving them would be a genuine advance independent of this paper.

Problem 8.2 (Virtual package realization)**.** *For each actual intermediate virtual scale, construct an actual compact address space, nonempty compact connected components realizing it, and exact polynomial branch, return, parent, sheet, critical-address, escape, localization, and root-enlargement data—without using any downstream width, compactness, rigidity, or MLC conclusion. The present paper solves the one-scale direct-decoration case; the virtual case at deeper copies would pass through a disconnected intermediate satellite object and currently assumes the virtual renormalization being constructed.*

Problem 8.3 (Marking transfer to the ends)**.** *Transfer a second satellite portrait to actual puzzle or generalized polynomial-like bases and actual iterates on the finite cylinders and end fibers of Theorem 5.3. Prove equivariance under (5.5), analytic realization at every end, the declared critical chronology, complete access and obstacle marking, terminal-return compatibility (compatibility with the separate root arrow τ_{root} of Remark 5.7 and the terminal cycle of Proposition 2.1; cf. Non-claims 2.3(N1)), and naturality under every authorized root enlargement (Theorem 5.8 and Corollary 6.7).*

8.3. **Relation to classical compactifications.** The address space (5.3) is the standard end compactification of the rooted binary tree, and nested-disk models of it are classical. The reader should therefore locate the contribution precisely: it is not the topology of Ω_∂ but the statements proved *about the dynamical realization*—membership in $K(f)$ and, under (H3), in $J(f)$; the exact component structure with its continuous closed component quotient; upper semicontinuity; the exact two-branch covering realization of the partial shift; the two-parent-map typing with a separate root arrow; fullness and nonseparation with the Cantor-indexed end compactum; the equivariant cell atlas with cellwise replacement at arbitrary addresses; and the finite protected carrier—together with the naturality of the compactification package under the root enlargement that the virtual program actually performs (Theorem 5.8 and Corollary 6.7). Each of these is used as a correctness criterion (a "positive control") against which a proposed virtual package can be audited field by field, and Appendix B records the boundary cases that audit must survive.

Machine-assistance disclosure. Drafting, verification, and build tooling for this paper: Fae Build, Anthropic Fable 5, and OpenAI GPT-5.6. All mathematical claims and their proofs are the author's responsibility.

Appendix A. Concordance with the program manuscript

This paper extracts, makes self-contained, and strengthens a package proved inside a larger program manuscript (*A Marked-Width Reduction of Mandelbrot Local Connectivity*, in preparation). For the reader with access to that manuscript, the concordance is:

This paper	**Program manuscript**
Theorem 5.3	Theorem 10.37
Theorem 5.4	— (new)
Theorem 5.5	— (new)
Theorem 5.8	Theorem 10.38 (strengthened)
Lemma 3.2	Lemma 10.39
Theorem 6.1	Theorem 10.40
Corollary 6.2	— (new)
Lemma 3.4	Lemma 10.41
Theorem 6.4	Theorem 10.42
Theorem 6.5	Theorem 10.43 (finite case; arbitrary F new)
Corollary 6.7	— (new)
Lemma 7.1	Lemma 10.44
Theorem 7.2	Theorem 10.45 (exact trace new)
Theorem 4.2	Theorem 10.4
Theorem 4.3	Theorem 10.8
Lemma 3.1	Lemma 10.10
Lemma 2.4	— (new)
Proposition 2.1	Proposition 10.31
Proposition 8.1	Proposition 10.32

The manuscript numbering refers to its Section 10 (protected regulated connectors and the direct-decoration pullback compactification).

Appendix B. Adversarial boundary cases

Each test below names a boundary case an auditor should try, and where the present proofs meet it.

Test B.1 (Closed pullbacks touching)**.** Can Q_0, Q_1 meet at a precritical boundary point? Distinctness of the open pullback components is insufficient in general—closed puzzle pullbacks may touch [KL08]. The exclusion here is exact: any common point lies on the depth-$(nq+1)$ puzzle graph, is regular for g because the critical orbit avoids the puzzle graphs, and Lemma 2.4 then bounds it to the closure of one component. Remark 2.6 records the deeper disjoint-Q^v fallback.

Test B.2 (Overlapping prime-end impressions)**.** Two distinct protected prime ends of a non-locally-connected fiber may have overlapping impressions. No statement in this paper requires disjoint closed tails: the certificate (Definition 4.1), the survival lemma (Lemma 3.7), and Step 1 of Theorem 7.2 use only pairwise disjoint *relative* collar tails $\overline{D_n} \setminus A$, which may have closures meeting inside the atom.

Test B.3 (Empty protected family)**.** With no protected chains, $P_{\mathrm{prot}} = \varnothing$ and Lemma 7.1 still produces the port (its proof branches on this case); Theorem 7.2 then reduces to the port and connector construction with $S = K_\epsilon$.

Test B.4 (Ancestor and descendant both selected)**.** Take $F = \{0, 001\}$. The cells D_0 and D_{001} are disjoint even though the addresses are comparable (Theorem 6.4, the ancestor–descendant clause), the packets stay in their cells, and the carrier avoids the descendant's end material because $Z \supset \bigcup_\omega E_\omega$.

Test B.5 (Infinite replacement family)**.** Take $F = \{0^k 1 : k \geq 0\}$, or $F = \mathtt{A}^{<\mathbb{N}}$. Theorem 6.5 gives compactness, the exact component structure, the continuous quotient, and (for full replacements) nonseparation: each depth-m approximant is finite regardless of $|F|$.

Test B.6 (Nonfull replacement)**.** Let one L_w be a compact connected set with a hole (say a circle in $\operatorname{int} D_w$ around K_w). Part (i) of Theorem 6.5 still holds—compactum, components, quotient—while $\widehat{\mathbb{C}} \setminus \widetilde{\mathcal{H}}$ is disconnected: the hole of L_w is a bounded complementary component. This separates the topological theorem from the fullness theorem.

Test B.7 (Root enlargement touching a child)**.** If $S \cap Q_a \neq \varnothing$, the separation Lemma 5.1(c) fails for the enlarged root, and ancestor–descendant fibers need no longer be disjoint. The first-level-gap hypothesis in (5.8) is therefore sharp for the present proofs.

Test B.8 (Root enlargement reaching the separator boundary)**.** If S meets ∂P but stays in $P \setminus (Q_0 \cup Q_1)$, the topological package survives (Theorem 5.8 uses only containment in P and disjointness from the children), but the cell atlas and fullness theory require $S \Subset G_0$, as in Corollary 6.7; the corollary's hypothesis is not cosmetic.

Test B.9 (Nondegenerate end fibers)**.** No proof in this paper uses shrinking cylinder diameters or assumes an end fiber is a point. Every argument—compactness, components, quotient, covering, fullness, cells, replacement, carrier—is valid for nondegenerate continuum end fibers.

Test B.10 (No lower semicontinuity)**.** The realization is upper semicontinuous, and π_∂ is continuous; neither implies that every point of E_ω is a limit of points of the finite fibers $K_{\omega|m}$. Lower semicontinuity is not proved and is used nowhere.

Test B.11 (Root arrow returning to the package)**.** Nothing proved here rules out $g(K_\epsilon) \cap \mathcal{H}_\partial \neq \varnothing$. The root arrow is separate typed data (Remark 5.7), and no permanent-escape statement enters any proof (Non-claims 2.3(N1)).

Test B.12 (Sphere extension moving infinity)**.** An arbitrary Schoenflies extension of ϕ_w need not commute with Fill: a sphere homeomorphism moving ∞ can exchange bounded and unbounded complementary components. Step 2 of Theorem 7.2 uses the pointed extension of Lemma 3.3, which fixes ∞ and does commute with Fill.

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